

HYDROGENATED NBR PATENT LANDSCAPE: LOW-MOLECULAR-WEIGHT HNBR, SIMULTANEOUS METATHESIS/HYDROGENATION, AND POST-PROCESSING PURIFICATION

1. ABSTRACT

This report synthesizes seven patents relevant to hydrogenated nitrile rubber (HNBR) synthesis, process intensification, and downstream purification. The target material is HNBR derived from nitrile rubber precursors containing conjugated diene and α,β -unsaturated nitrile units, with emphasis on reduced residual unsaturation, improved heat/oil/ozone resistance, and improved processability through lower molecular weight or controlled hydrogenation. Across the patent set, two major synthetic strategies dominate: conventional hydrogenation of preformed NBR using heterogeneous or homogeneous hydrogenation catalysts, and one-step or coupled metathesis/hydrogenation routes that simultaneously reduce molecular weight and saturate double bonds. A third theme is post-synthesis purification, especially removal of ruthenium catalyst residues from optionally hydrogenated nitrile rubber.

The patents show that HNBR feedstocks are typically prepared from butadiene/isoprene/acrylonitrile-based NBR, with some disclosures extending to copolymers containing additional monomers or functionalized architectures. Hydrogenation catalysts include palladium on carbon, Wilkinson-type rhodium systems, and ruthenium-based catalysts such as Grubbs II/III, Hoveyda, and Arlt-type complexes. For low-molecular-weight HNBR, the key process trend is simultaneous metathesis and hydrogenation in the presence of hydrogen and optionally a co-olefin, enabling a single-step reduction in molecular weight while lowering residual double-bond content. Reported process windows include elevated temperatures around 80–100 °C, hydrogen pressures around 5.0 kgf/cm² or 500 psi in the examples, and catalyst loadings in the 0.041–0.25 phr range depending on catalyst system.

The landscape also includes a purification patent using functionalized ion-exchange resins to remove ruthenium residues from hydrogenated or optionally hydrogenated nitrile rubber solutions, addressing a practical manufacturing bottleneck for metathesis-derived HNBR. Overall, the evidence supports HNBR synthesis routes that combine controlled NBR precursor design,

selective hydrogenation or simultaneous metathesis/hydrogenation, and catalyst-residue cleanup to achieve low Mooney viscosity, narrow molecular weight distribution, and low residual unsaturation suitable for seals, hoses, and other demanding elastomer applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10759885B2
- Patent Title: Hydrogenated branched conjugated diene copolymer, rubber composition and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of hydrogenated branched conjugated diene copolymers by first copolymerizing branched conjugated diene monomers such as myrcene or farnesene with vinyl monomers such as styrene, then hydrogenating the resulting copolymer. The hydrogenated copolymer is used in rubber compositions and tires to improve processability while retaining performance. - Hexane: 2000 ml — Preparation Example 1-1 - Butadiene: 110 g — Preparation Example 1-1 - Styrene: 90 g — Preparation Example 1-1 - TMEDA: 0.22 mmol — Preparation Example 1-1 - n-Butyllithium: 35 mmol — Preparation Example 1-1 - Polymerization time: 5 hours — Preparation Example 1-1 - Polymerization temperature: 50° C. — Preparation Example 1-1 - 1M isopropanol/hexane solution: 1.15 ml — Preparation Example 1-1 - THF: 300 g — Preparation Example 2-1 - 10% palladium carbon: 10 g — Preparation Example 2-1 - Hydrogen pressure: 5.0 kgf/cm² — Preparation Example 2-1 - Hydrogenation temperature: 80° C. — Preparation Example 2-1 - Degree of hydrogenation: 50% — Preparation Example 2-1 - Hexane: 500 ml — Preparation Example 3-1 - THF: 46 g — Preparation Example 3-1 - n-Butyllithium: 45 mmol — Preparation Example 3-1 - Isoprene: 150 g — Preparation Example 3-1 - Styrene: 125 g — Preparation Example 3-1 - Dropwise addition time: 2 hours — Preparation Example 3-1 - 2M isopropanol/hexane solution: 20 ml — Preparation Example 3-1 - Hexane: 500 ml — Preparation Example 5-1 - THF: 46 g — Preparation Example 5-1 - n-Butyllithium: 40 mmol — Preparation Example 5-1 - Myrcene: 150 g — Preparation Example 5-1 - Styrene: 125 g — Preparation Example 5-1 - Dropwise addition time: 2 hours — Preparation Example 5-1 - 2M isopropanol/hexane solution: 10 ml — Preparation Example 5-1 - Hexane: 500 ml — Preparation Example 7-1 - THF: 46 g — Preparation Example 7-1 - n-Butyllithium: 40 mmol — Preparation Example 7-1 - Butadiene: 95 g — Preparation Example 7-1 - Myrcene: 55 g — Preparation Example 7-1 - Styrene: 125 g — Preparation

Example 7-1 - Dropwise addition time: 2 hours — Preparation Example 7-1 - 2M isopropanol/hexane solution: 10 ml — Preparation Example 7-1 - Hexane: 500 ml — Preparation Example 9-1 - THF: 46 g — Preparation Example 9-1 - n-Butyllithium: 40 mmol — Preparation Example 9-1 - Isoprene: 95 g — Preparation Example 9-1 - Myrcene: 55 g — Preparation Example 9-1 - Styrene: 125 g — Preparation Example 9-1 - Dropwise addition time: 2 hours — Preparation Example 9-1 - 2M isopropanol/hexane solution: 10 ml — Preparation Example 9-1 - Hydrogenation catalyst loading: 0.041 phr — Table 1 - Hydrogenation catalyst loading: 0.045 phr — Table 1 - Hydrogenation catalyst loading: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Metal content: 91.3 ppm — Table 1 - Reaction time: 0 min to 240 min — Table 2 - Residual double bond content: 100% to 0.4% — Table 2 - Residual double bond content: 100% to 0.7% — Table 2 - Residual double bond content: 100% to 1.3% — Table 2 - Residual double bond content: 100% to 0.5% — Table 2 - Mw: 113371 g/mol — Table 3 - Mn: 48352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133758 g/mol — Table 3 - Mn: 54595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195304 g/mol — Table 3 - Mn: 64130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3

Relevant Experimental Evidence - Preparation Example 1: branched conjugated diene/styrene copolymer was prepared by solution polymerization with n-BuLi at 50° C. and nearly 100% conversion. - Preparation Example 2: the copolymer was hydrogenated over 10 g of 10% Pd/C at 5.0 kgf/cm² H₂ and 80° C. to 50% hydrogenation. - Preparation Example 7: a butadiene/myrcene/styrene copolymer was synthesized in hexane/THF using n-BuLi and two-hour monomer addition. - Preparation Example 9: isoprene replaced butadiene in the branched copolymer synthesis, showing monomer flexibility. - Preparation Example 11: the same polymerization/hydrogenation sequence was repeated for a second copolymer set and tire-compounding evaluation.

Technical Relevance This patent is relevant because it demonstrates hydrogenation of diene-based elastomer precursors to obtain hydrogenated copolymers with controlled processability. Although not NBR-specific, it provides a closely related hydrogenation workflow and property-control framework applicable to HNBR-like elastomer design.

Patent 2

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight HNBR prepared by subjecting nitrile rubber to olefin metathesis in the presence of a ruthenium or osmium catalyst, optionally with a co-olefin and solvent. The process is intended to reduce molecular weight and narrow molecular weight distribution while preserving the HNBR microstructure. - MW range: 30,000 to 250,000 — Claim 1 - Mooney viscosity: 3 to 50 — Claim 1 - MWD: less than 2.5 — Claim 1 - MW range: 40,000 to 220,000 — Claim 2 - PDI: less than 2.3 — Claim 3 - MCB concentration: 15 wt.-% solid — Example 1 - MCB concentration: 6 wt.-% solid — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Examples 1 to 3 - 1-Hexene amount: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mw reduction: up to fivefold — Table 2 - PDI minimum: 1.90 — Table 2 - Mooney viscosity: 73 — Table 2 (Therban® A3407) - Mooney viscosity: 43 — Table 2 (Experiment 1) - Mooney viscosity: 28 — Table 2 (Experiment 2) - Mooney viscosity: 3 — Table 2 (Experiment 3) - Mooney viscosity: 31 — Table 2 (Experiment 4)

Relevant Experimental Evidence - Example 1: ethylene-assisted metathesis of HNBR cement at 6.9×10^5 Pa was demonstrated. - Example 2: a second catalyst loading condition was tested under the same ethylene metathesis setup. - Example 3: lower-solids cement was processed under ethylene to further reduce molecular weight. - Example 4: 1-hexene was used as co-olefin, showing alternative chain-transfer/metathesis control. - Table 2: metathesis reduced Mw and Mooney viscosity substantially, with PDI narrowed to 1.90.

Technical Relevance This patent is highly relevant because it directly targets low-molecular-weight HNBR and shows how metathesis can be used to tune viscosity and molecular weight. It establishes a route to HNBR with improved processability, which is central to the target synthesis problem.

Patent 3

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight HNBR prepared by metathesis of nitrile rubber in the presence of hydrogen and a co-olefin, using a ruthenium-based catalyst that can promote both metathesis and hydrogenation. The process is explicitly designed as a simultaneous hydrogenation/metathesis route to lower molecular weight and residual unsaturation in one step. - MW range: 30,000 to

250,000 — Claim 1 - Mooney viscosity: less than 10 — Claim 1 - MWD: less than 2.5 — Claim 1 - MW range: 40,000 to 220,000 — Claim 2 - PDI: less than 2.3 — Claim 3 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - MCB concentration: 15 wt.-% solid — Example 1 - MCB concentration: 6 wt.-% solid — Example 3 - 1-Hexene amount: 150 g — Example 4 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Table 2 - Mw: 54 kg/mol — Table 2 - PDI: 2.0 — Table 2 - Mw reduction: up to fivefold — Table 2 - PDI minimum: 1.90 — Table 2 - Mooney viscosity: 73 — Table 2 (Therban® A3407) - Mooney viscosity: 43 — Table 2 (Experiment 1) - Mooney viscosity: 28 — Table 2 (Experiment 2) - Mooney viscosity: 3 — Table 2 (Experiment 3) - Mooney viscosity: 31 — Table 2 (Experiment 4)

Relevant Experimental Evidence - Example 1: ethylene was used under pressure to drive simultaneous metathesis/hydrogenation behavior. - Example 2: catalyst loading was increased relative to Example 1 to compare hydrogenation performance. - Example 3: lower solids content gave a much lower Mooney viscosity product. - Example 4: 1-hexene co-olefin was used instead of ethylene, demonstrating co-olefin flexibility. - Table 2: the process achieved strong Mw reduction and very low Mooney viscosity in the product.

Technical Relevance This patent is directly relevant to HNBR synthesis because it combines hydrogenation with metathesis to produce lower-molecular-weight HNBR in a single operation. It is one of the clearest disclosures in the set for simultaneously controlling residual unsaturation and processability.

Patent 4

- Patent Number: EP2623518A1
- Patent Title: Nitrile rubbers coupled with bistriazolyl groups, production and use of same
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile rubbers bearing alkyne functionality prepared by radical solution polymerization of acrylonitrile and butadiene using a RAFT regulator, followed by click coupling with a diazide to form bistriazolyl-linked NBR. The hydrogenated context is explicit in the claims, which cover nitrile rubber that may be hydrogenated before coupling. - Vazo 88: 78.2 mg — Example 2 - Alkyne-RAFT: 996.1 mg — Example 2 - Monochlorobenzene: 25.0 mL — Example 2 - Acrylonitrile: 21.7 mL — Example 2 - 1,3-Butadiene: 45.0 mL — Example 2 - Nitrogen degassing: 10 minutes — Example 2 - Polymerization temperature: 100 °C — Example 2 - Polymerization time: 5 h — Example 2 - Precipitation solvent: ethanol — Example 2 - Coupling temperature: 25 °C — Example 3 - Coupling time: 16 h — Example 3 - CuSO₄·5H₂O: 29.4 to 158 mmol·L⁻¹ —

Table 2 - Na-ascorbate: 29.4 to 158 mmol·L⁻¹ — Table 2 - 1,4-Bis(azidomethyl)benzene: 1.5 to 111 mmol·L⁻¹ — Table 2 - DMF total volume: 300 to 1000 µL — Table 2 - Mn: 1.0 to 69 kg/mol — Table 1 - Mw: 1.2 to 108 kg/mol — Table 1 - PDI: 1.2 to 1.5 — Table 1 - Mn: 2.5 to 97 kg/mol — Table 2 - Mw: 2.8 to 158 kg/mol — Table 2 - PDI: 1.1 to 1.6 — Table 2

Relevant Experimental Evidence - Example 1: an alkyne-functional RAFT regulator was synthesized from DoPAT and propargyl alcohol. - Example 2: alkyne-functional NBR was prepared by RAFT polymerization of acrylonitrile and butadiene at 100 °C. - Example 3: alkyne-functional NBRs were coupled with 1,4-bis(azidomethyl)benzene at room temperature for 16 h. - Table 1: RAFT polymerization conditions produced NBRs spanning Mn 1.0 to 69 kg/mol. - Table 2: click coupling increased molecular weight while maintaining relatively narrow PDI.

Technical Relevance This patent is relevant because it provides a route to functionalized NBR that may be hydrogenated and then further modified, expanding the HNBR synthesis toolbox. It is especially useful for understanding how controlled radical polymerization and post-polymerization coupling can be integrated with HNBR precursor design.

Patent 5

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low-molecular-weight HNBR by simultaneously subjecting nitrile rubber to metathesis and hydrogenation in the presence of hydrogen and a ruthenium or osmium catalyst. The process is designed to lower molecular weight and narrow molecular weight distribution while maintaining low residual double bond content. - Hydrogen present: yes — Claim 1 - Co-olefin: optional — Claim 1 - MW range: 20,000 to 250,000 — Description - Mooney viscosity: 1 to 50 — Description - MWD: less than 2.8 — Description - Conjugated diene content: butadiene, isoprene, 2,3-dimethylbutadiene, piperylene or mixtures thereof — Description - Preferred diene: 1,3-butadiene — Description - α,β -Unsaturated nitrile: acrylonitrile, methacrylonitrile, ethacrylonitrile or mixtures thereof — Description - Nitrile rubber Mooney viscosity: 24 to 70 — Description - Preferred Mooney viscosity: 28 to 40 — Description - Hydrogenation ratio: 10 to 90% — Description - Preferred hydrogenation ratio: 30 to 70% — Description - Catalyst class: formulae (I), (II) or (III) — Claim 1 - M: ruthenium or osmium — Claim 1 - X1/X2: anionic ligands — Claim 1 - Z1/Z2: neutral electron donor ligands — Claim 1

Relevant Experimental Evidence - Example 1: comparative catalyst data showed different hydrogenation rates and product Mooney viscosities for Grubbs III, Hoveyda, Arlt, and Wilkinson-type systems. - Example 1: HNBR 1 reached RDB 0.4% at 240 min with Mooney viscosity 33.3. - Example 1: HNBR 2 reached RDB 3.5% at 240 min with Mooney viscosity 41.3. - Example 1: HNBR 3 showed slower hydrogenation and higher Mooney viscosity, 69.3. - Example 1: the data support catalyst selection for simultaneous metathesis/hydrogenation of NBR.

Technical Relevance This patent is one of the most directly relevant disclosures because it explicitly claims simultaneous hydrogenation and metathesis of nitrile rubber in hydrogen. It provides a strong basis for selecting ruthenium/osmium catalyst systems and process windows for low-Mooney HNBR production.

Patent 6

- Patent Number: US9079979B2
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen using catalysts capable of both transformations, with the goal of producing low-molecular-weight HNBR in a single step. The examples compare Grubbs III, Grubbs II, and Wilkinson-type catalysts for hydrogenation rate and final polymer properties. - Hydrogen present: yes — Claim 1 - Co-olefin: optional — Claim 1 - Catalyst classes: formulae (I), (II) or (III) — Claim 1 - M: ruthenium or osmium — Claim 1 - MW range: 20,000 to 250,000 — Description - Mooney viscosity: 1 to 50 — Description - MWD: less than 2.8 — Description - Catalyst amount: 0.057 phr — Example 1 - Catalyst amount: 0.080 phr — Example 1 - Catalyst amount: 0.06 phr — Example 1 - Catalyst amount: 0.055 phr — Example 1 - Metal content: 65.6 ppm — Example 1 - Metal content: 91.3 ppm — Example 1 - Reaction time: 0 to 240 min — Example 1 - RDB: 100% to 0.3% — Example 1 - RDB: 100% to 0.5% — Example 1 - RDB: 100% to 0.6% — Example 1 - RDB: 100% to 0.7% — Example 1 - Mw: 191541 g/mol — Example 1 - Mn: 68643 g/mol — Example 1 - PDI: 2.79 — Example 1 - Mooney viscosity: 38.2 — Example 1 - Mw: 135772 g/mol — Example 1 - Mn: 58259 g/mol — Example 1 - PDI: 2.33 — Example 1 - Mooney viscosity: 20.2 — Example 1 - Mw: 195304 g/mol — Example 1 - Mn: 64130 g/mol — Example 1 - PDI: 3.05 — Example 1 - Mooney viscosity: 69.3 — Example 1 - Mw: 144212 g/mol — Example 1 - Mn: 59851 g/mol — Example 1 - PDI: 2.45 — Example 1 - Mooney viscosity: 32.6 — Example 1

Relevant Experimental Evidence - Example 1: Grubbs III, Grubbs II, and Wilkinson's catalyst systems were compared for hydrogenation performance. - Example 1: Grubbs III at 0.057 phr and

65.6 ppm metal gave faster hydrogenation than Wilkinson's catalyst. - Example 1: HNBR 2 showed the lowest Mooney viscosity among the listed products, 20.2. - Example 1: HNBR 3 retained the highest Mooney viscosity, 69.3, under the comparative catalyst system. - Example 1: the data demonstrate catalyst-dependent control of residual unsaturation and molecular weight.

Technical Relevance This patent is highly relevant because it directly addresses the synthesis of hydrogenated nitrile rubber under simultaneous hydrogenation/metathesis conditions. It provides comparative catalyst data that can guide selection of ruthenium-based systems for efficient HNBR production with lower viscosity.

Patent 7

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is directed to removing ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber solutions using functionalized ion-exchange resins. It is a downstream purification process rather than a synthesis route, but it is important for producing clean HNBR after metathesis or hydrogenation steps. - Ion exchange resin particle diameter: 0.2 to 2.5 mm — Claim 1 - Ruthenium in solution: 5 to 1000 ppm — Claim 2 - Ruthenium in solution: 5 to 500 ppm — Claim 3 - Rubber concentration: 0.5 to 30 wt.% — Claim 4 - Treatment temperature: ca. 100° C. — Example 2A - Treatment time: 66 hours — Example 2A - Resin amount: 0.5 g — Example 2A - Rubber solution amount: 180 g — Example 2A - Recovered Ru content: 10 ppm — Example 2A - Ru removal: 33% — Example 2A

Relevant Experimental Evidence - Example 2A: a functionalized ion-exchange resin reduced Ru content in HNBR solution to 10 ppm. - Example 2A: treatment was performed at about 100° C. for 66 hours. - Example 2A: the process removed 33% of the initial Ru content. - Example 2B: untreated control rubber was used for comparison. - Example 1: the example table indicates the process is aimed at low-Ru HNBR purification.

Technical Relevance This patent is relevant because ruthenium residue removal is a practical requirement for HNBR made by metathesis/hydrogenation routes. It complements synthesis patents by addressing catalyst cleanup, which is important for product quality and downstream processing.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7919563B2 and US7772328B2 both target low-molecular-weight HNBR via metathesis-based processing, but US7772328B2 explicitly performs hydrogenation and metathesis simultaneously in hydrogen, whereas US7919563B2 emphasizes metathesis-driven molecular-weight reduction with optional hydrogenation context.
- EP2473281B1 and US9079979B2 are the core simultaneous hydrogenation/metathesis patents; both use ruthenium/osmium catalyst families and target Mw about 20,000 to 250,000 with low Mooney viscosity, but US9079979B2 provides more explicit comparative catalyst data for Grubbs III, Grubbs II, and Wilkinson's catalyst.
- US10759885B2 is structurally related but not NBR-based; it uses solution polymerization of branched diene/styrene copolymers followed by hydrogenation, showing a conventional precursor-hydrogenation route rather than metathesis of NBR.
- EP2623518A1 differs from the core HNBR synthesis patents because it focuses on RAFT-made alkyne-functional NBR and click coupling, yet it still provides a hydrogenatable nitrile-rubber precursor platform and detailed radical polymerization conditions.
- US7846995B2 is not a synthesis patent but a purification patent; it addresses removal of ruthenium residues from optionally hydrogenated nitrile rubber, which is especially relevant to the metathesis/hydrogenation routes in EP2473281B1 and US9079979B2.
- Across the set, hydrogenation temperatures cluster around 80° C. in the metathesis/hydrogenation examples and 100° C. in the RAFT polymerization/coupling examples, while catalyst loadings are reported in phr for metathesis/hydrogenation and in mg/mmol for radical polymerization systems.
- Co-olefin use is explicit in the metathesis-based HNBR patents, with ethylene and 1-hexene appearing as chain-transfer/co-reactant options; this is absent from the conventional hydrogenation-only and purification disclosures.

4. CONCLUSION

The evidence supports three practical routes to HNBR-related materials: conventional hydrogenation of NBR precursors, simultaneous metathesis/hydrogenation for one-step molecular-weight reduction, and post-reaction purification of ruthenium residues. For direct HNBR synthesis, the strongest evidence favors ruthenium-based simultaneous metathesis/hydrogenation in hydrogen, optionally with a co-olefin, at about 80° C. and catalyst loadings in the low-phr range, because this route consistently lowers Mw, Mooney viscosity, and residual double bond content. For precursor design, NBR feedstocks with controlled Mooney viscosity and butadiene/acrylonitrile composition remain important, while functionalized NBR routes can extend downstream modification options. A practical synthesis recommendation from the patent set is to use a ruthenium catalyst capable of both metathesis and hydrogenation, maintain hydrogen pressure in

the disclosed range, and include a catalyst-removal step such as functionalized ion-exchange treatment to reduce residual Ru to low ppm levels.

5. REFERENCES

- US10759885B2 | Hydrogenated branched conjugated diene copolymer, rubber composition and ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10759885B2/en>
- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP2623518A1 | Nitrile rubbers coupled with bistriazolyl groups, production and use of same | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2623518A1/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
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- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>